

9 Plants store biomass both above and below ground.

- (a) Scientists have found that on average, 24 % of plant biomass is underground in the roots.

The total plant biomass underground contains 113 gigatonnes of carbon. This is equivalent to the mass of carbon dioxide emissions produced by humans in 10 years.

One gigatonne is 1×10^{12} kg.

Calculate the mass of carbon dioxide, in kilograms, produced by humans in one year.

Give your answer in standard form.

(2)

Answer kg

- (b) Describe how GALP, produced in the leaves, becomes incorporated into biomass in the roots.

(4)

[illegible]

(c) Inorganic ions are used by plants to make molecules.

The table shows some molecules made by plants.

For each molecule, put one cross ☒ in the appropriate box, in each row, to show which inorganic ion provides an atom found in the molecule.

(3)

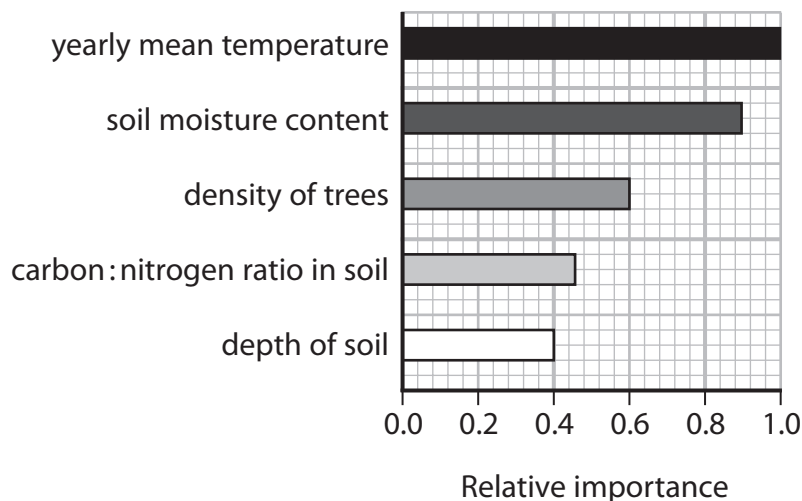
Molecule	Inorganic ion			
	nitrate only	phosphate only	both nitrate and phosphate	neither nitrate nor phosphate
cellulose	☒	☒	☒	☒
nucleic acid	☒	☒	☒	☒
triglyceride	☒	☒	☒	☒

*(d) Forests, shrublands and grasslands are three types of ecosystem.

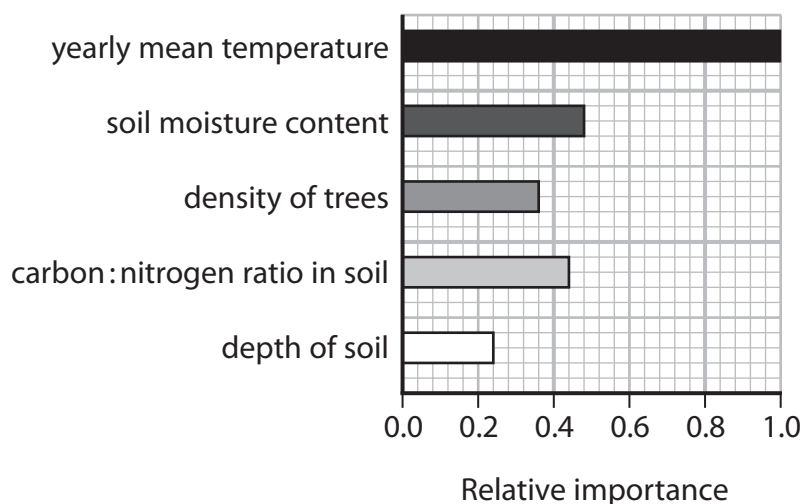
Biotic and abiotic factors affect how much biomass is stored in the roots of plants.

The graphs show the importance of some abiotic factors in determining how much biomass is stored in the roots in these ecosystems.

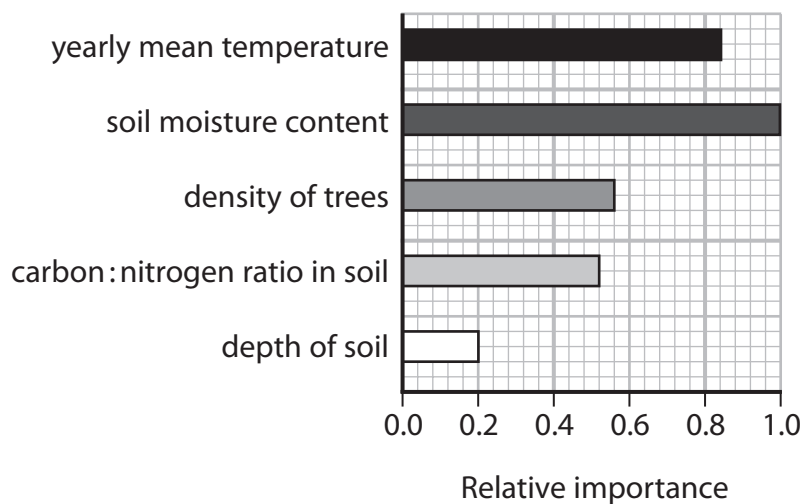
Forests



Shrublands



Grasslands



Discuss how these factors could affect the proportion of biomass stored in the roots of plants growing in these three ecosystems.

(6)

(Total for Question 9 = 15 marks)

TOTAL FOR PAPER = 90 MARKS